

Mobile Healthcare Application

One-page repo summary generated from local project evidence on 2026-03-25.

What It Is

An Expo/React Native mobile app for a healthcare monitoring workflow tied to AWS Amplify services and ESP32 firmware.

The implemented app supports auth, role-based admin/doctor views, patient assignment, and a placeholder for incoming pH or vitals data.

Who It's For

Primary users are doctors and admins. Doctors review assigned patients; admins create patients and reassign them to doctors.

How To Run

- From the repo root, go to 'frontend'.
- Run 'npm install'.
- Run 'npx expo start' or 'npm start'.
- Open the app in Expo Go, an emulator, simulator, or web.
- Backend deployment steps: Not found in repo.
- Firmware flashing steps: Not found in repo.

What It Does

- Lets users sign up, confirm email, and sign in with AWS Cognito via Amplify Auth.
- Routes users by role so admins land on the dashboard and doctors land in doctor tabs.
- Stores user profiles with name, email, role, and created date.
- Lets admins create patients, assign a doctor, and list all patients.
- Lets admins reassign a patient to a different doctor.
- Lets doctors refresh and view only their assigned patients.
- Shows a patient detail screen reserved for future pH or vitals readings.

How It Works

- Frontend: Expo Router screens in 'frontend/app' configure Amplify and call 'MedtronicHealthAPI' through 'frontend/src/api.ts'.
- Auth: Amplify uses a Cognito user pool and identity pool in 'us-east-2'; role checks read Cognito groups from the ID token.
- API: API Gateway routes '/users', '/patients', and '/doctors' to Lambda 'UserMeHandler'.
- Backend: 'UserMeHandler' runs an Express app that writes profiles to DynamoDB table 'Users-dev' and patient records plus links to 'Medtronic_Health'.
- Data model: patient creation and reassignment use DynamoDB transactional writes; doctor and admin views query the same table differently.
- Device side: ESP32 firmware reads UART data, converts it to a JSON payload with 'Voltage' and 'pH Value', then notifies over BLE.
- Mobile BLE ingestion path from the app to the firmware: Not found in repo.